Neurocomputing 101 (2013) 32–42

Contents lists available at SciVerse ScienceDirect

Neurocomputing

journal homepage: www.elsevier.com/locate/neucom

Divergence-based feature selection for separate classes

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ARTICLE INFO

Article history:
Received 12 September 2011
Received in revised form
31 March 2012
Accepted 7 June 2012
Communicated by Y. Chang
Available online 24 August 2012

Keywords:
Feature selection
Kullback–Leibler divergence
Separate classes
Relevance
Redundancy

ABSTRACT

Feature selection is one of the core issues in designing pattern recognition systems and has attracted considerable attention in the literature. Most of the feature selection methods in the literature only handle relevance and redundancy analysis from the point of view of the whole class, which neglects the relation of features and the separate classes. In this paper, we propose a novel feature selection framework to explicitly handle the relevance and redundancy analysis for each class label. Then we propose two simple and effective feature selection algorithms based on this framework and Kullback–Leibler divergence. An empirical study is conducted to evaluate the efficiency and effectiveness of our algorithms comparing with five representative feature selection algorithms. Empirical results show that our proposed algorithms are efficient and outperform the selected algorithms in most cases, and show the superiority of our proposed feature selection framework.

1. Introduction

Feature selection (variable selection or attribute selection) for classification modeling has attracted significant attention during the past decades both in pattern recognition and in machine learning. In the context of pattern recognition application, for example, the datasets we encounter may be characterized by high dimensional feature spaces, not all of these features are relevant and some are redundant. Dimension reduction is needed here not only for improving the efficiency of the classifier, but reducing the cost of feature acquisition in the system. Feature selection has proven in both theory and practice effective in enhancing learning efficiency, increasing classification accuracy, and reducing complexity of learned results [1].

Feature selection is the problem of selecting a subset from the original set of features in order to improve the quality of the data. Roughly, feature selection methods can be divided into three types [2,3], embedded methods, filters and wrappers. In embedded methods, feature selection is integrated into a given learning algorithm and performs in the training process, like C4.5 algorithm [4] does. Wrappers rely on performance estimated by one type of classifiers to select features and to evaluate the quality of those features. In contrast, filters select features based on some classifier agnostic criterions [2]. Those metrics that filter methods employ to estimate the discrimination capability generally include Fisher score [5], χ^2-test [6], mutual information [7–9], Symmetrical Uncertainty (SU) [1], etc. The main characteristic of wrapper methods is that they are sometimes computationally expensive in learning because they have to train and test the classifier for each feature subset candidate. Additionally, the performance of wrapper and imbedded methods are optimized directly by the selected learning method(s). Our study is mainly interested in developing fast filter methods.

Regarding the evaluation strategy, feature selection methods typically fall into two categories: feature ranking and feature subset selection. The former generally evaluates one feature at a time and outputs ranked features weighted by their individual predictive power [10–12], ignoring feature interaction and dependencies which may lead to redundancy. The latter evaluates feature subsets and searches for a best feature subset both considering individual predictive power and redundancy [1,13–16]. Owing to that an exhaustive search of 2^N-1 candidate subsets (where N is the number of features) is computationally intractable [17], a tradeoff between the computational cost and the quality of the generated feature subset have been taken under consideration in designing feasible feature selection methods [12,18,14,1,19]. Those intermediate methods usually employ heuristic search strategies (e.g., forward/backward search, sequential floating search [20], Best-First search [21]), to obtain a sub-optimal solution.

Most of the filter methods in the literature consider the relevance between features and the class from an average view. That is, they evaluate the predictive power of features by calculating the average (mean) score on different class labels, which neglects the relationship between the finally selected

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http://dx.doi.org/10.1016/j.neucom.2012.06.036

features and each class label), thus may leading to elimination of those "particularly relevant with a certain class label" features, and finally resulting in the lower performance of classification. We use an example to illustrate this.

Example 1.1. Given a dataset (the detail of which is shown in Table 1) containing 2 features F_1, F_2 and a class C with two labels {0,1}. From Table 1, it is obvious that $F_1=1$ corresponds well with $C=0$, while $F_2=0$ corresponds well with $C=1$, and hence F_1 and F_2 are both the strongly relevant features for the class C. However, for the typical feature selection algorithm FCBF [1], $SU(F_1,C)=0.420$ is greater than $SU(F_2,C)=0.375$, which means that FCBF determines F_1 a more relevant feature than F_2. In addition, $SU(F_1,F_2)=0.434$ is greater than $SU(F_1,C)=0.420$. According to the approximate Markov blanket strategy, FCBF will output $\{F_1\}$ as the selected feature subset. Unfortunately, the classification accuracy of Support Vector Machine (SVM) for $\{F_1\}$ is only 81.82%, which is much less than 90.91%—the classification accuracy of SVM for $\{F_1,F_2\}$.

Since feature selection is to select features that lead to a higher separability between the different classes, it is necessary to select features that can capture the individual characteristic of each class label and the skew in the class distribution, particularly for those imbalanced classes. Inspired by this, we propose in this paper a new feature subset selection framework and two feature selection methods. For the toy problem shown in Example 1.1, our methods respectively estimate the "distance" between F_1, F_2 and class label $C=0$, $C=1$. Taking one of the proposed algorithms *w* DFSC which we will introduce in Section 4 as an example, for $C=0$, we have $\mathrm{Diff}(F_1,C=0)=0.397>\mathrm{Diff}(F_2,C=0)=0.335$, and for $C=1$, we have $\mathrm{Diff}(F_2,C=1)=0.295>\mathrm{Diff}(F_1,C=1)=0.242$, hence *w* DFSC can successfully identify the most relevant feature for each class label (F_1 for $C=0$ and F_2 for $C=1$). The unique contributions that distinguish our work from existing literature are twofold:

- A novel feature selection framework is proposed to explicitly handle the relevance and redundancy analysis for each class label.
- Simple and effective divergence-based feature selection methods are developed based on the framework with comparative studies.

The remainder of the paper is organized as follows. We first propose in Section 3 a novel feature selection framework to explicitly select features for different class labels. Then we provide in Section 4 two new feature selection methods based on this framework and show the implementations in Section 5. In Section 6, we report and discuss the experimental results of the proposed algorithms in terms of efficiency and effectiveness comparing with the state-of-the-art algorithms on various well known datasets. Finally, in Section 7, we summarize the concluding remarks and point out the future work.

2. Related work

Roughly, we can classify the most representative feature subset selection techniques into two categories. One simultaneously considers the relevance between the feature subset and the class and the redundancy among features in the set and scores candidate subsets by a unique criterion, while the other decouples relevance and redundancy analysis and explicitly realizes them in two separate processes, and allows an individual search way in finding a subset rather than subset search. We call these two schemes one-step scheme and two-step scheme, respectively.

Fig. 1 shows the one-step scheme which is typically applied in Correlation based Feature Selection (CFS) algorithm proposed by Hall [13]. CFS employs $\mathrm{cor}=kr_{ic}/\sqrt{k+k(k-1)r_{ij}}$ to evaluate the quality of feature subsets, where k is the number of features in the current subset $\mathbf{S}$, r_{ic} is the average correlation between feature f_i ($f_i\in\mathbf{S}$) and the class C, r_{ij} is the average inter-correlation between f_i and f_j ($f_i,f_j\in\mathbf{S}, i\neq j$). CFS selects the subset which maximizes cor value as the output. The drawback of CFS is that it pays no attention to search approaches, which makes CFS have to combine with a certain search approach (e.g., Best-First search and sequential forward search) to select features. Ding and Peng [22] and Peng et al. [7] applied the analogous scheme, which they called mRMR criterion, to select minimal Redundancy but Maximal Relevance (mRMR) features. They defined redundancy measure $R=1/|\mathbf{S}|^2\sum_{x_i,x_j\in\mathbf{S}}I(x_i,x_j)$ and relevance measure $D=\sum_{x_i\in\mathbf{S}}I(x_i,C)$, where I denotes mutual information, and applied $\max\Phi(D,R)$, which takes the forms of $\max(D-R)$ or $\max(D/R)$, as the subset evaluation function. Analogous forms can also be

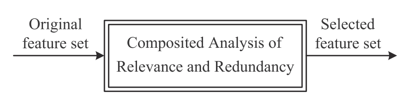
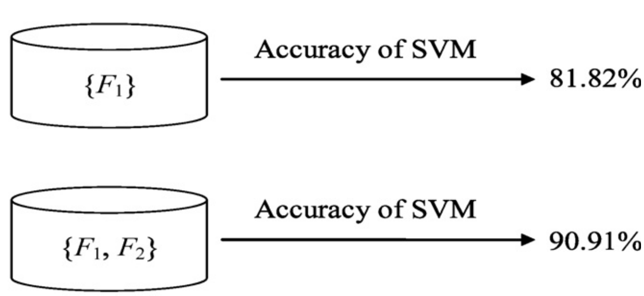


Fig. 1. The one-step scheme for feature ranking methods.

Table 1
A toy dataset.

F_1	F_2	C
1	1	0
1	2	0
1	3	0
1	4	0
1	5	0
0	0	1
1	0	1
2	0	1
3	0	1
4	2	1
5	1	1



discovered in [12,18], where they respectively defined their own D and R while both adopted $\max(D-\beta R)$ as the evaluation criterion with a parameter β to regulate relative significance of redundancy.

Apart from the above composite metrics, Conditional Mutual Information (CMI) is an existing metric in information theory which gives attention to both relevance and redundancy and hence is often used in feature selection. Flueret [14] and Wang et al. [23] harnessed Conditional Mutual Information Maximin (CMIM) criterion (in Flueret [14] it is also called Conditional Mutual Information Maximization) to select features under the one-step scheme. Given the current feature subset **S**, they argued the larger the value of $I(f;C|\tilde{f})$ ($\tilde{f} \in \mathbf{S}$), the higher quality the candidate feature f. That is, f carries available additional information about the class C, which does not have been caught by any $\tilde{f} \in \mathbf{S}$. To satisfy this, feature f of which the minimal $I(f;C|\tilde{f})$ for every $\tilde{f} \in \mathbf{S}$ is larger than that of any other feature should be picked into the current set. CMIM is based on the assumption that dependencies can be captured by only a pair of features, which makes CMIM ignore the dependencies among three or larger families of features in the feature subset **S**. Sotoca and Pla [24] utilized hierarchical clustering to provide a minimal Relevant Redundancy (mRR) selected feature subset which best compresses the discriminative information contained in the original set of features. They clustered features into several clusters using a CMI-based distance metric, and then gathered the center of each cluster together as the selected subset. The motivation is that the smaller the value of $\sum_i I(f_i;C|f_c)$, the better representation (more relevance with the class and less redundancy with features in its segment) the feature f_c in a certain segment, where f_i and f_c are members of the segment and $f_i \neq f_c$. Unfortunately, mRR also encounters the problems that CMIM does.

Although it is theoretically feasible that methods under the one-step scheme may achieve optimal solutions, they may suffer from a computational problem caused by searching through all of the feature subsets. In addition, it is likely that two or more candidate subsets are with similar evaluation score. When encountering this, the methods will fall into a dilemma of judging discrimination capabilities of these candidate subsets.

To solve the problems mentioned above many individually consider the two indices (relevance between feature and the class and redundancy among features) during their search procedures. Its advantage lies in that by analyzing relevance and redundancy individually, it avoids difficulties in subset search and allows more efficient and effective ways in finding an approximate optimal subset. Most of these methods apply the scheme in which relevance is considered first and redundancy later, or both are repetitiously considered, which is shown in Fig. 2.

A typical application of the two-step scheme is a refinement of CFS called Fast Correlation Based Feature selection algorithm (FCBF) by Yu and Liu [1] to circumvent the drawback of CFS and to speed up the search process. FCBF applies the two-step scheme under Markov blanket theory [25], which is a distribution-based theory and is proposed for the definition of optimal feature subset by Koller and Sahami [26]. FCBF employs Symmetrical Uncertainty (SU) as the metric to describe the correlation between candidate feature f and the class C and sort features in a descending order according to their SU values. Then it eliminates redundant features via an approximate Markov blanket criterion. FCBF is simple and fairly easy to implement, and in which the approximation it applied does speed up the search process since it estimates relevance and redundancy from the view of individual, rather than estimating those for each subset. However, FCBF suffers from instability, for the naive heuristics it employs will not be appropriate under any circumstance. Huang and Chow [8] proposed MIIO and MISF criteria and employed them to deal with irrelevance and redundancy respectively for every current candidate feature with a simple incremental search approach. The method they proposed is called QMIFS-p method. Dissimilar with FCBF, QMIFS-p identifies relevance and redundancy in every iteration (dashed arrow shown in Fig. 2), which enhances the ability for identifying potential redundant features during the search process.

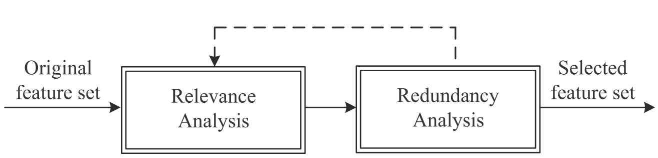


Fig. 2. The two-step scheme for feature subset selection methods.

It is noticed that series of Markov blanket discovery methods [27–30] which trend to induce the Markov blanket of a target depending on conditional independence test have also been applied to select optimal feature subset (see [15] for more details). The typical Markov blanket induction algorithms is the Incremental Association Markov Blanket (IAMB) algorithm [27], which contains growing and shrinking phases that well correspond to the two parts in the two-step scheme. The metric applied in IAMB for the conditional independence test is CMI. As mentioned earlier, CMI not only considers the information quantity of a feature about the class, but takes into account the redundancy between the feature and the conditioning set, which makes IAMB have the ability to avoid potential redundant features being selected in its growing phase and to further identify redundancy in its shrinking phase.

3. A new framework for feature subset selection

As mentioned above, most of the previous works focus on the identification of relation (relevance and redundancy) between features and the class, and apply statistical, information-theoretical or distance-based (and so on) metrics to evaluate the relation for the hardness to directly measure the similarity between two distributions. However, most of the metrics they used only score feature(s) on the 'average' level, but never concern the relation of feature(s) on each class label. Since the most important tasks for feature selection is to improve the classification performance, and the classification accuracy is finally reflected by both including the positive samples and excluding the negative ones for each class label, it is necessary for feature selection methods to evaluate relevance and redundancy on each class label. From this point of view, it is the feature that relevant/redundant to a certain class label rather than the class should be included in/be eliminated from the feature subset, which has inspired our new feature selection framework shown in Fig. 3.

The new framework shown in Fig. 3 is distinctly characterized by its explicitly handling feature selection task for each class label, and one can achieve the goal of efficiently find an optimal feature subset through the framework interpreted into two different processes: if the composite analysis of relevance and redundancy (shown in Fig. 1) is applied for each class label, one can simply collect the selected subset of features for each class label as the final selected subset or may evaluate those subsets iteratively (the dashed arrow shown in Fig. 3) and then determine the final feature subset. If relevance analysis and redundancy analysis (shown in Fig. 2) are applied separately, one can conduct the relevance analysis first to get relevant subsets for each class label and then conduct the redundancy analysis (the dashed arrow shown in Fig. 3), in such a way as to achieve the final selected subset. Algorithms

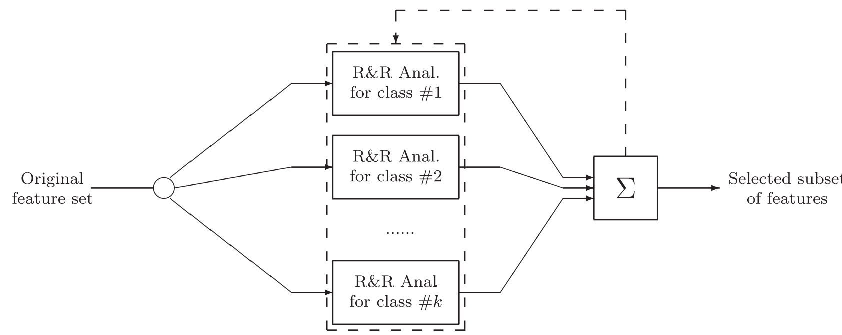


Fig. 3. A new framework of feature selection.

proposed in the next section belong to the latter pattern. It is obvious that this framework shows equivalent attention to all the class labels, whereas traditional methods only focus on the average score on the relation between candidate feature(s) and the class and hence may neglect the influence of some rare class labels during the selecting process. That is, for instance, a highly scored feature or feature subset may catch less information about some class labels which contains fewer samples than others for the number of those class labels is not significant and hence make little difference to the average score. In addition, since classification algorithms often faces the incompatible problem that improving the predictive accuracy of the rare classes slightly will be at the cost of seriously decreasing the accuracy of the large classes [31], it is important to select features that can capture the skew in the class distribution. From this point of view, dealing with each class label explicitly may provide a better solution of feature selection, especially for those unbalanced datasets.

4. Efficient feature selection methods based on the proposed framework

A data instance is typically described to the system as an assignment of values $\mathbf{f}=\{f_1,\ldots,f_n\}$ to a set of features $\mathbf{F}=\{F_1,\ldots,F_n\}$. Usually we assume that all of the data instances (samples) are drawn from some probability distribution over the space of feature vectors. Formally, for each assignment of values $\mathbf{f}$ to $\mathbf{F}$, we have a probability $P(\mathbf{F}=\mathbf{f})$. Classification is a procedure that takes as input an unlabeled sample and classifies it as belonging to one of a number of possible classes $c_1,\ldots,c_k$. The classifier must make the decision based on the assignment $\mathbf{f}$ associated with an unlabeled sample. Theoretically the feature vector will fully determine the appropriate classification for the unlabeled samples. However, it is often the case that we do not have enough features to make this a deterministic decision. Therefore, we use a probability distribution to model the classification function. More precisely, for each assignment of values $\mathbf{f}$ to $\mathbf{F}$ we have a distribution $P(C|\mathbf{F}=\mathbf{f})$ on different possible classes $C=\{c_1,\ldots,c_k\}$. A learning algorithm implicitly uses some approximate style of the conditional distribution $P(C|\mathbf{F})$ – for example, the empirical frequencies observed in the training set – to construct a classifier for the problem. It is well-known that the number of features has a strong effect on the performance of a learning algorithm. On the one hand, the existence of irrelevant or redundant features can degrade the accuracy of a learning algorithm, causing it to use less-than-optimal features for classification. On the other hand, the computational complexity of many learning algorithms depends heavily on the number of features. Therefore, it is often useful to reduce the irrelevant and redundant features before the classifier is constructed.

Formally, let p and q be two distributions over some probability space Ω. The Kullback–Leibler (KL) divergence of p and q is defined as

$$D_{\mathrm{KL}}(p\|q)=\sum_{x\in\Omega}p(x)\log\frac{p(x)}{q(x)}.$$

Note that the roles of p and q are not symmetrical in this definition. Generally speaking, the idea is that p is the original distribution, and q is an approximation candidate to it. Then, $D_{\mathrm{KL}}(p\|q)$ measures the extent of the similarity that we make using q as a substitute for p. Let $\mathbf{G}$ be some subset of $\mathbf{F}$, from [32] we know that a feature $F\in\mathbf{F}\backslash\mathbf{G}$ is relevant to the class C if $\exists c\in C, P(F|\mathbf{G},C=c)\neq P(F|\mathbf{G})$, which we denote as $P_c(F|\mathbf{G})\neq P(F|\mathbf{G})$. And $F\in\mathbf{F}\backslash\mathbf{G}$ is an irrelevant/redundant features given the feature subset $\mathbf{G}$ if $\forall c\in C, P_c(F|\mathbf{G})=P(F|\mathbf{G})$. Thus, we may apply a dynamic sequential forward strategy to search for a suitable subset. That is, for each current feature subset $\mathbf{G}$ and each class label c_i, we select the feature F most relevant to c_i. In order to have a metric which allows us to compare one feature F to another, we must integrate the values $D_{\mathrm{KL}}(P_{c_i}(F|\mathbf{f_G})\|P(F|\mathbf{f_G}))$ for different feature assignments $\mathbf{f_G}$ into a single quantity. Naively, we might think to simply sum the KL divergence for the different feature vectors, or to consider the maximum KL divergence over all feature vectors. Neither of these ideas take into consideration the fact that some feature vectors are far more likely to occur than others, and that we might not mind making a larger mistake in certain rare cases. Therefore, we apply

$$\mathrm{Diff}_{\mathbf{G}}(F,c_i)=\sum_{\mathbf{f}_G\in\mathbf{G}}P_{c_i}(\mathbf{f}_G)\cdot D_{\mathrm{KL}}(P_{c_i}(F|\mathbf{f}_G)\|P(F|\mathbf{f}_G))$$

to measure the relevance and the redundancy of feature F to the class label c_i ($c_i\in C$) given the feature subset $\mathbf{G}$. And we determine the feature F_i^+ satisfying

$$\max_{F\in\mathbf{F}\backslash\mathbf{G}}\mathrm{Diff}_{\mathbf{G}}(F,c_i)$$

to be the candidate relevant feature for each class label c_i under the current subset $\mathbf{G}$. As far as the sample insufficiency problem is concerned, it is proper to set an independent coefficient λ to justify whether a candidate feature is/is not a relevant or redundant feature, that is, a feature will be identified to be a relevant feature of the class label c_i if satisfying

$$\max_{F\in\mathbf{F}\backslash\mathbf{G}}\mathrm{Diff}_{\mathbf{G}}(F,c_i)\geq\lambda. \tag{1}$$

Then we update $\mathbf{G}=\mathbf{G}\cup\{F_i^+|i=1,2,\ldots,k\}$, and turn to the next iteration. If Eq. (1) does not hold for all the class labels in an iteration, the algorithm will stop searching and then output the selected feature subset $\mathbf{G}$.

When the imbalance of class distribution is considered, it is noted that the proper value of λ may change during the feature selection process for the sample size varies with different class label c_i. Small sample size may significantly reduce the reliability in independence (conditional independence) determination. Given two independent random variables X and Y with $\overline{X}=0$ and $\overline{Y}=0$, $|\overline{X}_n - \overline{Y}_n|$ converges in probability to 0 when n gets large ($n \to \infty$), where $\overline{X}_n$ and $\overline{Y}_n$ are the averages of the selected n samples of X and Y, respectively. From this point of view, we may conclude that the more samples we observe from two actual independent features F_i and F_j, the less relevant they are from our determination, and consequently we should set λ reasonably small when the sample size is large enough. For those with not enough samples, we appropriately set λ as a slightly larger number. To achieve this, we use $\delta_i = N/N_{c_i}$ as the weighted factor to adjust the value of λ for each class label, where N denotes the number of samples in the dataset, and N_{c_i} denotes the number of samples that belong to the class label c_i. That is, we apply

$$\max_{F \in \mathbf{F} \setminus \mathbf{G}} \mathrm{Diff}_{\mathbf{G}}(F,c_i) \geq \delta_i \lambda \tag{2}$$

as the criterion to identify relevant or irrelevant (redundant) features when taking the imbalance problem into consideration. We call the algorithm using Eq. (1) as the criterion Divergence-based Feature selection for Separate Classes algorithm, and the one using Eq. (2) weighted Divergence-based Feature selection for Separate Classes algorithm. We show the pseudo code of the algorithms below.

Algorithm 1. DFSC: Divergence-based Feature selection for Separate Classes (w DFSC: weighted-DFSC).

Input: dataset $\mathbf{D}$, with feature set $\mathbf{F}$ and class C.
Output: Selected feature subset $\mathbf{G}$
repeat
  Initialize $\mathbf{G} = \emptyset$, $TempSet = \emptyset$, $\delta_i = 1$ for all $i = 1, \ldots, |C|$
  (or $\delta_i = N/N_{c_i}$ for w DFSC).
  for each $F \in \mathbf{F} \setminus \mathbf{G}$ **do**
    for $i=1$ **to** $|C|$ **do**
      Find the feature F_i^+ **satisfying** $\max_{F \in \mathbf{F} \setminus \mathbf{G}} \mathrm{Diff}_{\mathbf{G}}(F,c_i) \geq \delta_i \lambda$
      $TempSet = TempSet \cup F_i^+$
    end for
    $\mathbf{G} = \mathbf{G} \cup TempSet$
    $TempSet = \emptyset$
  end for
until $\mathbf{G}$ is not changed

5. Estimation of $\mathrm{Diff}_{\mathbf{G}}(F,c_i)$

We describe in this section how to estimate $\mathrm{Diff}_{\mathbf{G}}(F,c_i)$ from a dataset $\mathbf{D}$ in the proposed algorithms. All the estimations we discuss here are implemented on discrete features. A certain discrete procedure is needed before the estimation when dealing with continuous or mixed datasets. For the sake of convenience, we denote $\mathrm{Diff}_{\mathbf{G}}(F,c_i)$ estimated on dataset $\mathbf{D}$ as $\mathrm{Diff}_{\mathbf{G},\mathbf{D}}(F,c_i)$.

For the simplest case $\mathbf{G} = \emptyset$, we have

$$\hat{\mathrm{Diff}}_{\emptyset,\mathbf{D}}(F,c_i) = \mathrm{D}_{\mathrm{KL}}(\hat{P}_{c_i}(F) \| \hat{P}(F)) = \sum_{f \in F} \frac{\pi_{c_i}(f)}{N_{c_i}} \log \frac{\frac{\pi_{c_i}(f)}{N_{c_i}}}{\frac{\pi(f)}{N}}, \tag{3}$$

where N is the whole sample size, N_{c_i} denotes # samples belonging to a certain class label c_i, $\pi(f)$ denotes # samples containing $F=f$, and $\pi_{c_i}(f)$ denotes # samples belonging to a certain class label c_i and containing $F=f$. Since # feature here is constant (only one feature F) and # values and # value assignments of (F,c_i) will never exceed the sample size N, there exists a linear solution for $\mathrm{Diff}_{\emptyset,\mathbf{D}}(F,c_i)$ estimation under the assumption that values (or value assignments) which do not appear in the data are improbable and their probability is approximated by zero in the absence of other information. Note that if we use vectors to record those intermediate parameters ($\pi_{c_i}(\cdot)$,N_{c_i}, etc.) indexed by the class label c_i, the estimations of $\mathrm{Diff}_{\emptyset,\mathbf{D}}(F,c_1)$,$\mathrm{Diff}_{\emptyset,\mathbf{D}}(F,c_2), \ldots ,\mathrm{Diff}_{\emptyset,\mathbf{D}}(F,c_k)$ can be totally conducted within $O(N)$ time, for the # value assignments of (F,C) also will never exceed the sample size N. Here we describe in detail how to estimate $\mathrm{Diff}_{\mathbf{G},\mathbf{D}}(F,c_i)$ with an nonempty set $\mathbf{G}$. We have

$$\begin{aligned}\hat{\mathrm{Diff}}_{\mathbf{G},\mathbf{D}}(F,c_i) &= \sum_{\mathbf{f_G} \in \mathbf{G}} \hat{P}_{c_i}(\mathbf{f_G}) \mathrm{D}_{\mathrm{KL}}(\hat{P}_{c_i}(F|\mathbf{f_G}) \| \hat{P}(F|\mathbf{f_G})) \\ &= \sum_{\mathbf{f}_G \in \mathbf{G}} \sum_{f \in F} \frac{\pi_{c_i \& \mathbf{f}_G}(f)}{N_{c_i}} \log \frac{\frac{\pi_{c_i \& \mathbf{f}_G}(f)}{N_{c_i \& \mathbf{f}_G}}}{\frac{\pi_{\mathbf{f}_G}(f)}{N_{\mathbf{f}_G}}} \\ &= \sum_{\mathbf{f}_G \in \mathbf{G}} \sum_{f \in F} \frac{N_{c_i \& \mathbf{f}_G}}{N_{c_i}} \left[\frac{\pi_{c_i \& \mathbf{f}_G}(f)}{N_{c_i \& \mathbf{f}_G}} \log \frac{\frac{\pi_{c_i \& \mathbf{f}_G}(f)}{N_{c_i \& \mathbf{f}_G}}}{\frac{\pi_{\mathbf{f}_G}(f)}{N_{\mathbf{f}_G}}} \right],\end{aligned} \tag{4}$$

where $\pi_{\mathbf{f}_G}(f)$ and $N_{\mathbf{f}_G}$ denotes # samples containing $F=f$ and # samples on the data subset of which each sample contains the value assignment $\mathbf{f}_G$ respectively, and $\pi_{c_i \& \mathbf{f}_G}(f)$ and $N_{c_i \& \mathbf{f}_G}$ denotes # samples containing $F=f$ and # samples on the data subset of which each sample contains the label c_i and the value assignment $\mathbf{f}_G$ respectively. Denoting the samples containing $\mathbf{f}_G$ as a dataset $\mathbf{D}_{f_G}$, we have

$$\hat{\mathrm{Diff}}_{\emptyset,\mathbf{D}_{f_G}}(F,c_i) = \sum_{f \in F} \frac{\pi_{c_i}(f)}{N_{c_i}} \log \frac{\frac{\pi_{c_i}(f)}{N}}{\frac{\pi(f)}{N}}. \tag{5}$$

From the point of view of dataset $\mathbf{D}$, Eq. (5) can be rewritten as

$$\hat{\mathrm{Diff}}_{\emptyset,\mathbf{D}_{f_G}}(F,c_i) = \sum_{f \in F} \frac{\pi_{c_i \& \mathbf{f}_G}(f)}{N_{c_i \& \mathbf{f}_G}} \log \frac{\frac{\pi_{c_i \& \mathbf{f}_G}(f)}{N_{c_i \& \mathbf{f}_G}}}{\frac{\pi_{\mathbf{f}_G}(f)}{N_{\mathbf{f}_G}}}. \tag{6}$$

Let $N_{c_i \& \mathbf{f}_G}/N_{c_i} = \eta_{\mathbf{f}_G}$. With Eqs. (4) and (6), we get

$$\hat{\mathrm{Diff}}_{\mathbf{G},\mathbf{D}}(F,c_i) = \sum_{\mathbf{f}_G \in \mathbf{G}} \eta_{\mathbf{f}_G} \cdot \hat{\mathrm{Diff}}_{\emptyset,\mathbf{D}_{f_G}}(F,c_i). \tag{7}$$

Eq. (7) shows a relation between $\mathrm{Diff}_{\mathbf{G},\mathbf{D}}(F,c_i)$ and $\mathrm{Diff}_{\emptyset,\mathbf{D}}(F,c_i)$. If the data subsets $\mathbf{D}_{f_G}$ – hereafter we denote them as $\mathbf{D}_1, \mathbf{D}_2, \ldots, \mathbf{D}_n$, $\bigcup_{i=1}^n \mathbf{D}_i = \mathbf{D}$, $n \leq N$, and n is the number of value assignments of the features in $\mathbf{G}$ – are separated from each other, the estimation of $\mathrm{Diff}_{\emptyset,\mathbf{D}_i}(F,c_i)$ can be conducted within $O(N_{\mathbf{D}_i})$ time, and hence the estimation of $\mathrm{Diff}_{\mathbf{G},\mathbf{D}}(F,c_i)$ can be conducted within $\sum_{i=1}^n O(N_{\mathbf{D}_i}) = O(N)$ time. Also, if we apply vectors to record those intermediate parameters indexed by the class label, we can estimate all the $\mathrm{Diff}_{\mathbf{G},\mathbf{D}}(F,c_i)$, $i = 1,2,\ldots,k$ within $O(N)$ time. Since DFSC and w DFSC always add new features into the subset $\mathbf{G}$ after all the estimations of $\mathrm{Diff}_{\mathbf{G},\mathbf{D}}(F,c_i)$ have been done, we may locally update the order of the samples to keep the data subset $\mathbf{D}_i \subset \mathbf{D}$ being continuously arranged in $\mathbf{D}$ without overlapping. To achieve this, the radix sort technique is applied in our algorithms. For the worst case k features being selected during an iteration, the sorting algorithm will update the dataset within $O(k(N+r))$ time, where r is the upper bound of the range of value for the features in the current subset $\mathbf{G}$. And hence the worst time complexity is $O(k(N+r)+N) = O(k(N+r))$ for the estimation of $\mathrm{Diff}_{\mathbf{G},\mathbf{D}}(F,c_i)$, and $O(k(N+r)|\mathbf{F}|^2)$ for the algorithms (both DFSC and w DFSC).

6. Empirical results

In this section, we empirically evaluate the performance of DFSC and w DFSC by comparing them with the state-of-the-art feature selection algorithms from both feature ranking and feature subset selection approaches. Algorithms from the former one are ReliefF [33] and Information Gain (IG) based feature ranking algorithm. ReliefF searches for nearest neighbors of samples for each class label and weights features in terms of how well they differentiate samples for different class labels. Algorithms from the latter one are IAMB [27], CFS [13] and FCBF [1]. For ReliefF, we use 5 neighbors and 30 instances throughout the experiments as suggested by [33]. For IAMB, we set the independent threshold $\eta = 0.01$ as suggested by [15]. For CFS, we apply Best-First as its search approach. For FCBF, we set the predefined threshold $\gamma = 0$ as suggested by [1]. And for the proposed DFSC and w DFSC when comparing them with feature subset selection methods, we heuristically set the threshold $\lambda = 0.01$. The experimental platform is Weka (Waikato environment for knowledge analysis) [34], which is an excellent tool in data mining and contains most of the state-of-the-art feature selection algorithms. The proposed DFSC and w DFSC algorithm are implemented in Java and can be applied in Weka platform. All experiments are conducted on a 64-bit Linux computer with a 2.40 GHz CPU, 4 GB RAM. Datasets and classifiers used in the validation process are described in the following sections.

6.1. Benchmark datasets

Twelve benchmark datasets frequently used in literatures are adopted in our experiments to fairly validate performance, where eight of them (Arrhythmia, kr-vs-kp, Musk2, Splice, Mushroom, DNA, Madelon and Gisette) are from the UCI Machine Learning Repository[1] and the rest are well known public challenge datasets.[2] Table 2 summarizes general information about these datasets. Note that these datasets differ greatly in the sample size (range from 203 to 72,626) and the number of features (range from 22 to 12,600), which can provide a comprehensive testing and suit for feature selection methods under different conditions. For continuous and mixed datasets, we take MDL method [35] to discrete continuous features before applying feature selection methods.

6.2. Classifiers for validation

In our experiments, four widely used classifiers, NBC (Naïve Bayesian Classifier) [34], k NN (k-Nearest Neighbor) [36], C4.5 [4], and SVM (Support Vector Machine) are chosen to evaluate prediction accuracy on the selected features. Brief descriptions about these classifiers are given as follows.

- Naïve Bayesian Classifier [34] is a simple probabilistic classifier with an assumption of conditional independence among the features, i.e., the presence (or absence) of a particular feature of a class is unrelated to the presence (or absence) of any other feature. It only requires a small amount of training data to estimate the parameters necessary for classification. Many experiments have demonstrated that NBC has worked quite well in many complex real-world situations and outperformed many other classifiers.
- k-Nearest Neighbor [36] is a supervised non-linear classifier. Given a parameter k and an sample d, the k NN technique considers the k training examples closest to d in terms of their distance (e.g., Euclidean distance) in the feature space, and then predicts the class of d as the dominant real class among those k neighbors by a majority voting manner. A value of $k=1$ neighbor has been used in our experiments.

Table 2
Description of datasets.

Datasets	# samples	# features	# classes
Arrhythmia	452	279	16
Lung Cancer	203	12,600	5
Splice	3190	61	3
Kr-vs-kp	3196	36	2
Musk2	6598	166	2
Mushroom	8124	22	2
DNA	3186	180	3
Madelon	2000	500	2
Gisette	6000	5000	2
Ibn-sina	20,722	92	2
Sylva	72,626	216	2
Hiva	21,339	1617	2

- C4.5 [4] is an extension of ID3 algorithm. It applies an information gain based criterion to classifier samples using decision tree which is built in a top-down manner. Each feature of the data can be used to make a decision that splits the original sample set into two smaller subsets. The algorithm examines the normalized information gain to make the decision. To decrease the level of over-fitting, the tree can be further pruned.
- Support Vector Machine [37] is a classifier that has become very popular for the last decade. It harnesses a kernel to transform the original data in a transformed space where a linear classifier is applied. In our paper, the LibSVM package [38] has been used, which can deal with both two-class and multi-class classification. The kernel used is a radial basis function (RBF). LibSVM provides well performance in RBF kernels, estimating the parameters g and c.

In order to increase the statistical significance of the results with a limited number of samples as well as to achieve impartial results among the feature subset selection methods, the average values over ten 10-fold cross validations have been used for each classifier in verifying classification capability. Student's paired two-tailed t-tests with a significance level of 0.01 between accuracies have been conducted in order to determine the statistical significance of the difference of the results when comparing with feature subset selection methods. For the experiments of feature ranking methods, we only report the average values over one 10-fold cross validation.

6.3. Comparison among feature subset selection methods

6.3.1. Results of selected features and execution time

Table 3 records the execution time for DFSC, w DFSC, and each selected feature subset selection algorithm. We can observe that CFS is the most time-consuming algorithm, while w DFSC is one of the most efficient algorithms on the whole. The average execution time of w DFSC is 5.164 s, shorter than all other feature subset selection algorithms. For the datasets with both small number of samples and features, the execution time for DFSC, w DFSC, IAMB, CFS, and FCBF is not with much variance. For the datasets with large number of samples (more than 5000 samples), w DFSC performs most efficiently among the five feature subset feature selection algorithms. For the typical feature subset selection algorithm CFS, however, the execution time on Lung Cancer (12,600 features) is much longer than other algorithms, which lies in that the feature subset searching strategy applied by CFS is not competent to deal with large scale datasets. The result of execution time verifies that w DFSC is comparatively efficient and can tackle the computationally intractable problem when dealing with high dimensional datasets.

[1] http://archive.ics.uci.edu/ml/.

[2] Ibn-sina, Sylva and Hiva are all available from http://www.causality.inf.ethz.ch/activelearning.php?page=datasets#cont, Lung Cancer are available from http://datam.i2r.a-star.edu.sg/datasets/krbd/LungCancer/LungCancer-Harvard1.html.

Table 4 records the number of features selected by DFSC, *w* DFSC and selected feature subset selection algorithms. We observe from the table that the average number of selected features of DFSC (14.4) and *w* DFSC (9.2) are smaller than most of the algorithm used in our experiment, and only a little bit larger than IAMB (8.1). Fig. 4 shows the comparison between each feature selection algorithm and the original set, from which we can see that DFSC, *w* DFSC, IAMB, CFS and FCBF all dramatically reduce the dimensionality by selecting only a small portion of the original features on those datasets.

Table 3
Execution time on selected datasets.

Datasets	Execution time (s)				
	DFSC	*w* DFSC	IAMB	CFS	FCBF
Arrhythmia	0.079	0.048	0.110	0.213	0.039
Lung Cancer	0.516	0.375	0.776	8592.049	16.850
Splice	0.119	0.072	0.114	0.157	0.123
Kr-vs-kp	0.104	0.046	0.054	0.184	0.234
Musk2	1.231	0.660	0.764	1.987	0.515
Mushroom	0.073	0.034	0.037	0.105	0.067
DNA	0.583	0.379	0.881	0.329	1.129
Madelon	1.151	0.801	0.737	0.960	0.200
Gisette	42.056	44.131	56.167	608.801	31.933
Ibn-sina	2.316	1.965	2.835	2.153	1.537
Sylva	9.462	0.573	1.888	5.252	2.039
Hiva	346.923	12.878	31.233	291.626	33.253
Avg.	33.718	5.164	7.967	791.985	7.327

Table 4
Number of selected features.

Datasets	# features				
	DFSC	*w* DFSC	IAMB	CFS	FCBF
Arrhythmia	30	21	17	20	8
Lung Cancer	8	7	5	288	453
Splice	12	9	9	6	22
Kr-vs-kp	16	10	11	3	7
Musk2	12	9	7	14	2
Mushroom	4	3	3	1	4
DNA	20	15	17	4	38
Madelon	11	9	9	6	2
Gisette	16	18	7	76	28
Ibn-sina	9	7	9	3	5
Sylva	20	1	2	4	7
Hiva	15	1	1	31	12
Avg.	14.4	9.2	8.1	38.0	49.0

6.3.2. Results of different classifiers

Tables 5–8 show the average accuracy of NBC, 1NN, C4.5 and SVM on twelve datasets over ten 10-fold cross validations, respectively. For each classifier, the accuracies of classification on datasets with original features are given in "Unselect" row. Notation "a" or "b" denotes that the result of classification with current selector is significantly better or worse than those without feature selection (i.e., the corresponding "Unselect" column) in Student's paired two-tailed *t*-test with a significant level 0.01, respectively. Bold value in each row shows the best classification result among five selectors. The last row "W/T/L" in each table

Table 5
Accuracy of NBC on selected features, which records ten 10-fold cross validation accuracy rate (%). The last row of this table (W/T/L) summarizes over all data sets the wins/ties/losses in accuracy (at significance level 0.01) comparing various feature subsets with full set.

Datasets	Unselect	DFSC	*w* DFSC	IAMB	CFS	FCBF
Arrhythmia	62.39	69.89[a]	**70.73**[a]	67.37[a]	69.47[a]	65.86[a]
Lung Cancer	89.06	92.17[a]	91.97[a]	87.98[b]	**95.62**[a]	95.57[a]
Splice	95.42	95.01	**96.90**[a]	90.68[b]	93.52[b]	96.16[a]
Kr-vs-kp	87.79	93.19[a]	91.24[a]	**94.26**[a]	90.43[a]	92.14[a]
Musk2	83.91	79.38[b]	83.92	76.20[b]	65.61[b]	**84.08**
Mushroom	95.76	98.64[a]	**98.86**[a]	**98.86**[a]	98.52[a]	98.52[a]
DNA	93.93	94.59[a]	**94.62**[a]	94.46[a]	86.50[b]	93.35[b]
Madelon	58.46	59.16[a]	59.13[a]	59.13[a]	**60.83**[a]	59.36[a]
Gisette	85.00	**89.34**[a]	89.28[a]	84.12[b]	87.59[a]	83.92[b]
Ibn-sina	73.37	87.95[a]	89.02[a]	88.76[a]	**90.61**[a]	90.11[a]
Sylva	94.20	97.82[a]	**99.17**[a]	97.15[a]	96.95[a]	98.87[a]
Hiva	85.70	95.20[a]	96.58[a]	96.58[a]	95.06[a]	**96.81**[a]
Avg.	83.75	87.70	**88.45**	86.30	85.89	87.90
W/T/L		10/1/1	11/1/0	8/0/4	9/0/3	9/1/2

[a] Significant win at 0.01 level over unselected results.
[b] Significant loss at 0.01 level over unselected results.

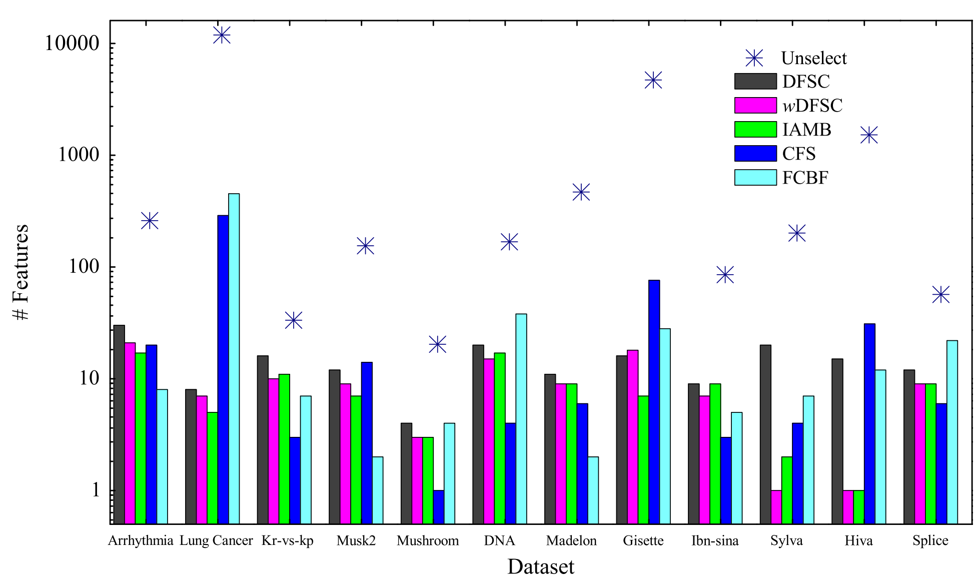


Fig. 4. Comparison of the number of selected features by DFSC. *w* DFSC, IAMB, CFS, and FCBF.

Table 6
Accuracy of 1NN on selected features, which records ten 10-fold cross validation accuracy rate (%). The last row of this table (W/T/L) summarizes over all data sets the wins/ties/losses in accuracy (at significance level 0.01) comparing various feature subsets with full set.

Datasets	Unselect	DFSC	*w* DFSC	IAMB	CFS	FCBF
Arrhythmia	53.21	59.23[a]	**64.19**[a]	54.91[a]	63.30[a]	59.82[a]
Lung Cancer	97.39	93.65[b]	**95.57**[b]	87.04[b]	95.47[b]	94.78[b]
Splice	74.43	84.71[a]	87.21[a]	81.62[a]	**90.02**[a]	81.21[a]
Kr-vs-kp	96.12	**98.15**[a]	97.38[a]	97.13	90.43[b]	94.18[b]
Musk2	94.68	**94.74**[a]	94.57	93.32[b]	93.01[b]	91.42[b]
Mushroom	100	**100**	99.70[b]	99.70[b]	98.52[b]	98.57[b]
DNA	74.55	90.59[a]	**92.39**[a]	90.64[a]	87.16[a]	83.13[a]
Madelon	55.10	**86.50**[a]	85.99[a]	85.99[a]	77.20[a]	56.10[a]
Gisette	79.91	**91.61**[a]	91.55[a]	81.15[a]	87.59[a]	83.73[a]
Ibn-sina	96.22	**96.36**[a]	96.13[b]	**95.15**[b]	92.55[b]	93.06[b]
Sylva	98.17	**99.78**[a]	99.08[a]	98.79[a]	98.76[a]	98.68[a]
Hiva	95.69	96.45[a]	96.58[a]	96.58[a]	**96.89**[a]	96.85[a]
Avg.	84.62	90.98	**91.70**	88.50	89.24	85.96
W/T/L		10/1/1	8/1/3	7/1/4	7/0/5	7/0/5

[a] Significant win at 0.01 level over unselected results.
[b] Significant loss at 0.01 level over unselected results.

Table 7
Accuracy of C4.5 on selected features, which records ten 10-fold cross validation accuracy rate (%). The last row of this table (W/T/L) summarizes over all data sets the wins/ties/losses in accuracy (at significance level 0.01) comparing various feature subsets with full set.

Datasets	Unselect	DFSC	*w* DFSC	IAMB	CFS	FCBF
Arrhythmia	65.64	68.03[a]	68.27[a]	65.31	68.94[a]	**70.09**[a]
Lung Cancer	90.44	92.27[a]	**92.51**[a]	88.52[b]	91.77[a]	89.80
Splice	94.17	**94.47**[a]	94.28	90.37[b]	90.02[b]	93.43[b]
Kr-vs-kp	99.44	**98.47**[b]	97.50[b]	97.26[b]	90.43[b]	94.03[b]
Musk2	96.84	**95.76**[b]	95.58[b]	94.55[b]	95.72[b]	91.22[b]
Mushroom	100	**100**	99.70[b]	99.70[b]	98.52[b]	98.62[b]
DNA	92.54	**94.60**[a]	94.47[a]	94.30[a]	87.16[b]	91.37[b]
Madelon	69.08	78.26[a]	**78.52**[a]	**78.52**[a]	73.31[a]	63.80[b]
Gisette	87.20	**92.95**[a]	92.92[a]	88.43[a]	87.15	83.67[b]
Ibn-sina	97.32	**96.96**[b]	96.91[b]	96.87[b]	93.87[b]	94.22[b]
Sylva	99.39	**99.71**[a]	99.17[b]	99.17[b]	99.11[b]	99.17[b]
Hiva	96.93	**97.00**[a]	96.58[b]	96.58[b]	96.82[b]	96.83[b]
Avg.	90.75	**92.37**	92.20	90.80	89.40	88.85
W/T/L		8/1/3	5/1/6	3/1/8	3/1/8	1/1/10

[a] Significant win at 0.01 level over unselected results.
[b] Significant loss at 0.01 level over unselected results.

summarizes the wins/ties/losses in accuracy comparing five feature subsets with original features over all datasets.

Results in Table 5 show the performance of Naïve Bayesian classifier using the five feature subset selection algorithms. We can see that in general all selectors either maintain or improve the accuracy of NBC, and *w* DFSC surpasses other five selectors in most cases. As an illustration, the W/T/L for *w* DFSC and DFSC are 11/1/0 and 10/1/1 respectively, which achieve largest quantities of cases with significant better performance and smallest quantities of cases with significant worse performance among five selectors. In addition, *w* DFSC achieves maximal values of accuracy five times, whereas most of the other selectors only achieve no more than twice, and own the highest average classification accuracy of 88.45. For DFSC, the average accuracy is 87.70, which is also larger than IAMB and CFS.

Results in Table 6 show the performance of k NN ($k=1$) classifier using the five feature subset selection algorithms. The performance of *w* DFSC is pronounced since the result of average accuracy for *w* DFSC is better than any other selectors. DFSC achieves best W/T/L (10/1/1), and the accuracy is 90.98, showing

Table 8
Accuracy of SVM on selected features, which records ten 10-fold cross validation accuracy rate (%). The last row of this table (W/T/L) summarizes over all data sets the wins/ties/losses in accuracy (at significance level 0.01) comparing various feature subsets with full set.

Datasets	Unselect	DFSC	*w* DFSC	IAMB	CFS	FCBF
Arrhythmia	54.20	54.20	54.20	54.20	54.20	**54.78**[a]
Lung Cancer	68.47	68.47	68.47	68.47	68.47	68.47
Splice	92.35	93.25[a]	93.25[a]	88.95[b]	**93.87**[a]	92.71
Kr-vs-kp	94.16	94.93[a]	**96.15**[a]	95.19[a]	90.43[b]	94.08[b]
Musk2	90.38	90.22	90.50	90.25	90.35	91.46[a]
Mushroom	100	**100**	99.70[b]	99.70[b]	98.52[b]	99.02[b]
DNA	95.51	**95.13**[b]	95.11[b]	94.90[b]	87.16[b]	92.65[b]
Madelon	50.00	**51.24**[a]	51.12[a]	50.01	50.08	51.07[a]
Gisette	50.00	61.21[a]	**65.94**[a]	63.19[a]	52.47[a]	60.48[a]
Ibn-sina	96.37	**96.57**[a]	95.94[b]	96.41	92.77[b]	92.43[b]
Sylva	99.17	99.16	99.17	**99.18**	99.17	99.15[b]
Hiva	96.55	**96.94**[a]	96.58	96.58	96.82[a]	96.70[a]
Avg.	82.26	83.44	**83.84**	83.09	81.19	82.75
W/T/L		6/5/1	4/5/3	2/7/3	3/5/4	5/2/5

[a] Significant win at 0.01 level over unselected results.
[b] Significant loss at 0.01 level over unselected results.

Table 9
A comparison of wins/ties/losses between DFSC and *w* DFSC, IAMB, CFS, FCBF.

Classifiers	*w* DFSC	IAMB	CFS	FCBF
NBC	1/5/6	6/3/3	7/2/3	4/3/6
1NN	6/1/5	10/1/1	8/0/4	9/1/2
C4.5	4/7/1	10/1/1	9/3/0	11/0/1
SVM	3/7/2	6/5/1	6/5/1	6/4/2

Table 10
A comparison of wins/ties/losses between *w* DFSC and DFSC, IAMB, CFS, FCBF.

Classifiers	DFSC	IAMB	CFS	FCBF
NBC	6/5/1	8/3/1	9/0/3	6/2/4
1NN	5/1/6	9/3/0	9/1/2	11/0/1
C4.5	1/7/4	7/5/0	9/2/1	9/1/2
SVM	2/7/3	6/5/1	6/4/2	7/2/3

that DFSC is close to *w* DFSC and is superior to other three selectors. Similar results can also be found for C4.5 and SVM that DFSC and *w* DFSC also hold the best results in both W/T/L and classification accuracy than other algorithms (see Tables 7 and 8). Note that *w* DFSC has better results than DFSC in NBC, but similar and inferior results in k NN and C4.5 respectively. A possible reason for this lies in that NBC is a probabilistic-based learning model while k NN and C4.5 is not. For those unbalanced datasets, *w* DFSC considers sample inefficiency problem and uses a probabilistic-based method to partially tackle the problem, and thus making *w* DFSC be inherently consistent with NBC model. For C4.5 however, *w* DFSC does not pay enough attention to the class label with large sample size since the weighted factor δ for this label is always greater than 1, and thus may finally leading to the reduction of classification accuracy.

We also show W/T/L results comparing DFSC and *w* DFSC with other three selectors respectively in Tables 9 and 10. Results shown in the tables clearly reveal that DFSC and *w* DFSC outperforms CFS and FCBF in most cases. And the best results are achieved by applying 1NN for both DFSC and *w* DFSC. For *w* DFSC, all the results are superior to IAMB, CFS, and FCBF, whereas for DFSC there exists an inferior result (the W/T/L is 4/3/6) when comparing with FCBF by applying NBC without considering *w*

DFSC. On the whole, *w* DFSC outperforms DFSC, particularly in NBC and 1NN classifiers.

6.4. Comparison among feature ranking methods

To illustrate the effect of selected features by our methods, we also show the results on six datasets (kr-vs-kp, DNA, Madelon, Ibn-sina, Sylva, and Hiva) using NBC, *k* NN, C4.5 and SVM classifiers with two representative feature ranking methods ReliefF and IG. Since DFSC and *w* DFSC are feature subset selection methods and the number of selected features changes only with respect to different values of the independent threshold λ, we show the results from the largest number of features when $\lambda = 0$ to the smallest number of features (often 1 feature) when λ rises to the critical number, i.e., λ could not be any larger or else the methods will choose no features. Δ_λ between any two steps is set as 0.00001. Note that the number of selected features of our methods may be nonconsecutive when λ is increased by Δ_λ. We report the average accuracy of four classifiers for each selector in Figs. 5–10. All the experimental results are achieved in the manner of 10-fold cross validation.

The results in Figs. 5–10 show that DFSC and *w* DFSC outperform other selectors in majority cases. More precisely, the superiority of DFSC and *w* DFSC is clearly revealed on kr-vs-kp, DNA, and Sylva datasets. For Madelon dataset, DFSC and *w* DFSC also achieve better results than ReliefF and IG when the number of selected features is more than 5, although neither of them can

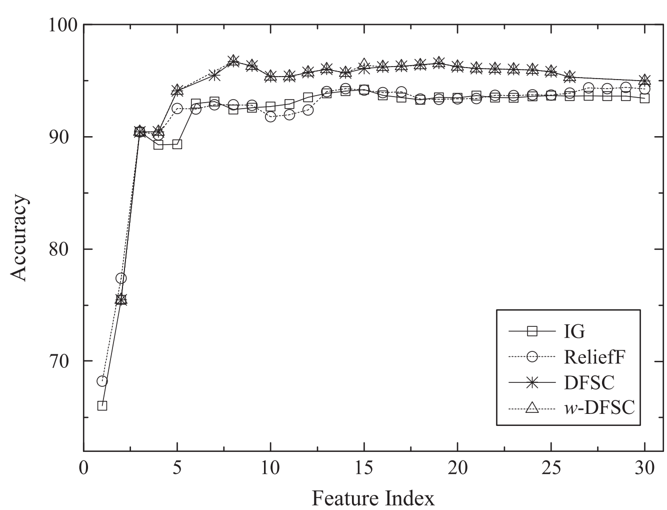


Fig. 5. Accuracy vs. different number of features on kr-vs-kp dataset.

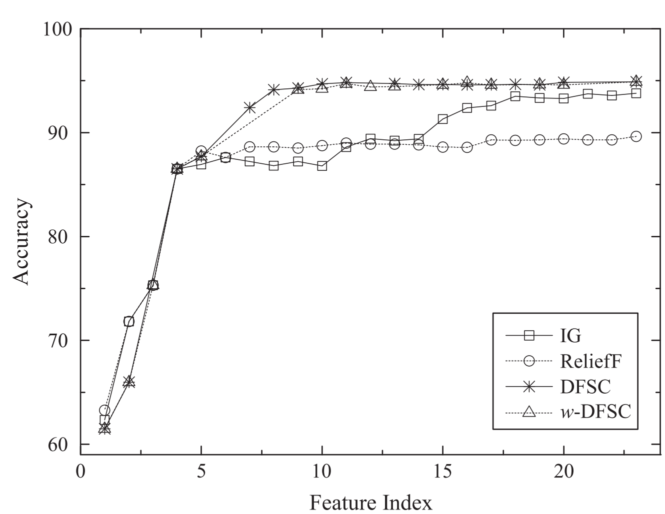


Fig. 6. Accuracy vs. different number of features on DNA dataset.

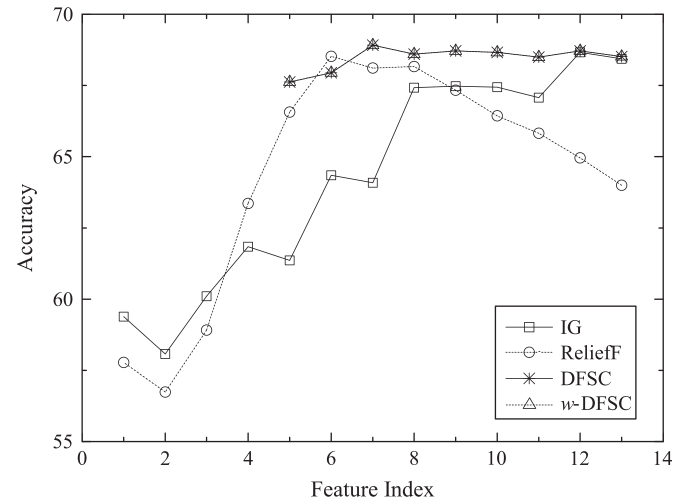


Fig. 7. Accuracy vs. different number of features on Madelon dataset.

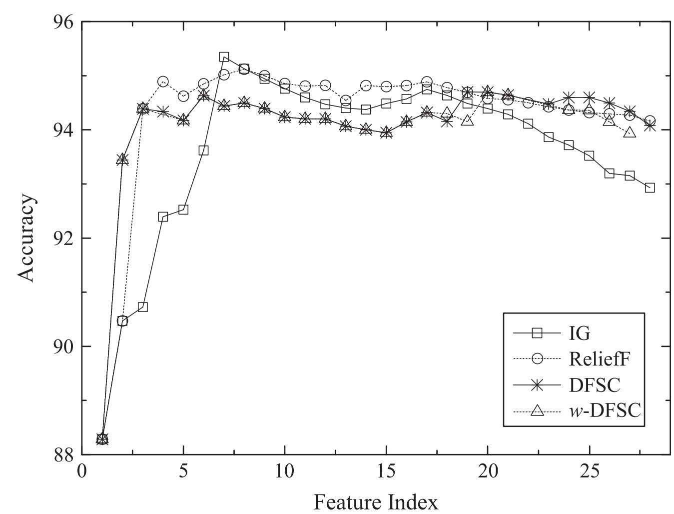


Fig. 8. Accuracy vs. different number of features on Ibn-sina dataset.

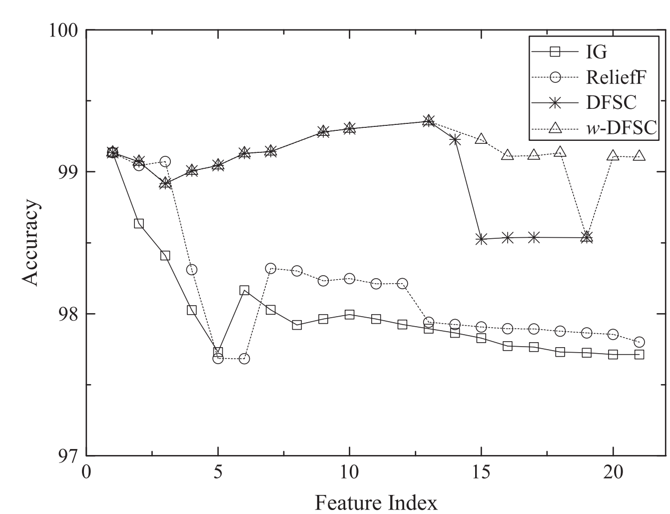


Fig. 9. Accuracy vs. different number of features on Sylva dataset.

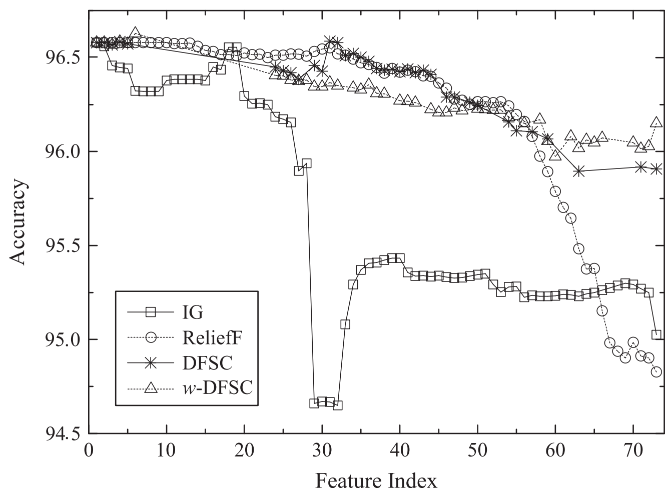


Fig. 10. Accuracy vs. different number of features on Hiva dataset.

select less than five features. For Ibn-sina dataset, however, the capabilities of DFSC and *w* DFSC are lower than ReliefF and IG when the number of selected features is less than 18. But when the number is more than 18, the results of DFSC and *w* DFSC are better than IG, and is nearly the same as that of ReliefF. The situation has turned over when the number of features surpasses 22. Note that the figures on Sylva and Hiva are different with those on the rest datasets. The possible reason behind this is that Sylva and Hiva may leads to the class label with the larger sample size strongly related to one or two features for they are extremely unbalanced datasets. In such a case redundancy appears when more than one features are selected, and thus directly leads to the lower performance of classification.

7. Conclusion

Since the classification accuracy is finally reflected by both including the positive samples and excluding the negative ones for each class label, we have proposed a novel feature selection framework to explicitly handle the relevance and redundancy analysis for different class labels. Simple and effective feature selection algorithms DFSC and *w* DFSC have been proposed based on the framework and evaluated through several experiments comparing with five representative and frequently used feature selection algorithms. Experimental results have shown that *w* DFSC as well as DFSC achieve more compact feature subset and better accuracy in classification, and have verified them to be comparable or better than those of state-of-the-art feature selection algorithms. We also have found that *w* DFSC outperforms DFSC on the whole, for the adjusted independence parameter $\delta_i\lambda$ may be more effective than λ for the classes with different sample size.

One of the factors that characterize DFSC and *w* DFSC is that they employ the estimation of joint distribution when calculating $\mathrm{Diff}_{\mathbf{G}}(F,c_i)$, which has been a barrier to designing feature subset selection strategies for two main reasons: one is that its computational requirements are generally enormous even with a medium-sized feature subset **S**, while the other is the sample insufficiency problem arising from real-life datasets. Note that the former one has been tackled in our work (see Section 5), whereas the latter one is still an issue – although we applied $\delta_i\lambda$ instead of λ in *w* DFSC to partially tackle this problem – that is worth being addressed in-depth by our feature work. Additionally, the satisfactory performance of *w* DFSC and DFSC are encouraging to further study many attractive issues such as online stream feature selection under our proposed feature selection framework.

Acknowledgments

We thank the editors and anonymous reviewers for their very useful comments and suggestions.

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